PROBABILITY AND STATISTICS

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Families of Continuous Random Variables

- **Conditioning a Continuous Random Variable:**
- *Definition: Conditional PDF given an Event*
- For a random variable X with PDF $f_X(x)$ and an event $B \subset S_X$ with $P[B] > 0$, the conditional PDF of X given B is

$$f_{X|B}(x) = \begin{cases} \dfrac{f_X(x)}{P[B]} & x \epsilon B \\ 0 & otherwise \end{cases}$$

- Theorem: Given an event space $\{B_i\}$ and the conditional PDFs $f_{X|B_i}(x)$,

$$f_X(x) = \sum_{all\ i} f_{X|B_i}(x) P[B_i]$$

Families of Continuous Random Variables

- **Definition:** *Conditional Expected Value Given an Event*
- If $\{x \in B\}$, the conditional expected value of X is

$$E[X|B] = \int_{-\infty}^{+\infty} x f_{X|B}(x)dx$$

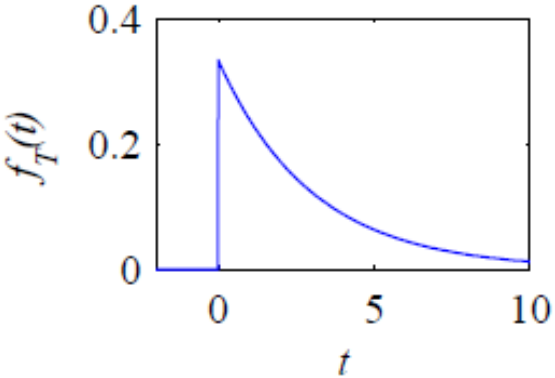
- To find the variance:

$$E[X^2|B] = \int_{-\infty}^{+\infty} x^2 f_{X|B}(x)dx$$

$$Var[X|B] = E[X^2|B] - E^2[X|B]$$

Families of Continuous Random Variables

- Example: Suppose the duration T (in minutes) of a telephone call is an exponential (1/3) random variable:



- $f_T(t) = \begin{cases} \frac{1}{3} e^{-t/3} & t \geq 0 \\ 0 & otherwise \end{cases}$

- For calls that last at least 2 minutes, what is the conditional PDF of the call duration? Find the expected value?

Families of Continuous Random Variables

- Example: The probability density function of random variable Y is

$$f_Y(y) = \begin{cases} \frac{1}{10} & 0 \le y < 10 \\ 0 & others \end{cases}$$

- Find the following:
- (1) P[Y ≤ 6]
- (2) the conditional PDF $f_{Y\,|Y\le 6}(y)$
- (3) P[Y > 8]
- (4) the conditional PDF $f_{Y\,|Y>8}(y)$
- (5) E[Y |Y ≤ 6]
- (6) E[Y |Y > 8]

Joint Random Variables

- **Joint Cumulative Distribution Function**
- **Definition:** Joint Cumulative Distribution Function (CDF)
- The joint cumulative distribution function of random variables X and Y is

$$F_{X,Y}(x,y) = P[X \leq x, Y \leq y]$$

- The joint CDF is a complete probability model.
- The joint CDF has properties that are direct consequences of the definition. For example, we note that the event $\{X \leq x\}$ suggests that Y can have any value so long as the condition on X is met. This corresponds to the joint event $\{X \leq x, Y < \infty\}$. Therefore,

$$F_X(x) = P[X \leq x] = P[X \leq x, Y < \infty] = \lim_{y \to \infty} F_{X,Y}(x,y) = F_{X,Y}(x,\infty)$$

- We obtain a similar result when we consider the event $\{Y \leq y\}$.

Joint Random Variables

- Theorem: For any pair of random variables, X, Y ,

1. $0 \leq F_{X,Y}(x, y) \leq 1$,
2. $F_X(x) = F_{X,Y}(x, +\infty)$
3. $F_Y(y) = F_{X,Y}(+\infty, y)$,
4. $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$,
5. If $x \leq x_1$ and $y \leq y_1$, then $F_{X,Y}(x, y) \leq F_{X,Y}(x_1, y_1)$,
6. $F_{X,Y}(+\infty, +\infty) = 1$

- **Joint Probability Mass Function**
- **Definition:** *Joint Probability Mass Function (PMF)*
- The joint probability mass function of discrete random variables X and Y is

$$P_{X,Y}(x, y) = P[X = x, Y = y]$$

Joint Random Variables

- Example: Test two integrated circuits one after the other. On each test, the possible outcomes are a (accept) and r (reject). Assume that all circuits are acceptable with probability 0.9 and that the outcomes of successive tests are independent. Count the number of acceptable circuits X and count the number of successful tests Y before you observe the first reject. (If both tests are successful, let Y = 2.) Draw a tree diagram for the experiment and find the joint PMF of X and Y.

Joint Random Variables

- **Theorem:** For discrete random variables X and Y and any set B in the X, Y plane, the probability of the event {(X, Y) ∈ B} is

$$P[B] = \sum_{(x,y)\in B} P_{X,Y}(x,y)$$

- **Example:** Continuing the previous Example, find the probability of the event B that X, the number of acceptable circuits, equals Y, the number of tests before observing the first failure.

- **Example:** The joint PMF $P_{Q,G}(q,g)$ for random variables Q and G is given in the following table:

$P_{Q,G}(q,g)$	$g=0$	$g=1$	$g=2$	$g=3$
$q=0$	0.06	0.18	0.24	0.12
$q=1$	0.04	0.12	0.16	0.08

- Calculate the following probabilities:
- (1) P[Q = 0] (2) P[Q = G]
- (3) P[G > 1] (4) P[G > Q]

Joint Random Variables

- **Marginal PMF**
- In an experiment that produces two random variables X and Y , it is always possible to consider one of the random variables, Y , and ignore the other one, X. In this case, we can use the methods of Chapter 2 to analyze the experiment and derive $P_Y(y)$, which contains the probability model for the random variable of interest.

- On the other hand, if we have already analyzed the experiment to derive the joint PMF $P_{X,Y}(x,y)$, it would be convenient to derive $P_Y(y)$ from $P_{X,Y}(x,y)$ without reexamining the details of the experiment.

- **Theorem:** For discrete random variables X and Y with joint PMF $P_{X,Y}(x,y)$,

$$P_X(x) = \sum_{y \in S_Y} P_{X,Y}(x,y)\,, \qquad P_Y(y) = \sum_{X \in S_X} P_{X,Y}(x,y)\,,$$

- It implies that we can find $P_Y(y)$ by summing $P_{X,Y}(x,y)$ over all points in $S_{X,Y}$ with the property $Y = y$. In the sum, y is a constant, and each term corresponds to a value of $x \in S_X$. Similarly, we can find $P_X(x)$ by summing $P_{X,Y}(x,y)$ over all points X, Y such that $X = x$.

Joint Random Variables

- Example: In first Example, we found the joint PMF of X and Y to be

$P_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.01	0	0
$x = 1$	0.09	0.09	0
$x = 2$	0	0	0.81

- Find the marginal PMFs for the random variables X and Y.

Joint Random Variables

- The most useful probability model of a pair of **continuous random** variables is a generalization of the PDF of a single random variable
- **Definition :** Joint Probability Density Function (PDF)
- The joint PDF of the continuous random variables X and Y is a function $f_{X,Y}(x,y)$ with the property

$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)dudv$$

- We know that $f_{X,Y}(x,y)$ is a derivative of the CDF

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}$$

Joint Random Variables

- **Theorem:**

$$P[x_1 \le X \le x_2, y_1 \le Y \le y_2]$$
$$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1)$$

- **Theorem:**
- A joint PDF $f_{X,Y}(x, y)$ has the following properties corresponding to first and second axioms of probability
- $f_{X,Y}(x, y) \ge 0 \;\; for\ all\ x, y$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x, y)dxdy = 1$

- **Theorem:** The probability that the continuous random variables (X, Y) are in A is

$$P[A] = \iint_A f_{X,Y}(x, y)dxdy$$

Joint Random Variables

- Example: Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c & 0 \le x \le 5, 0 \le y \le 3 \\ 0 & otherwise \end{cases}$$

- Find the constant c and $P[A] = P[2 \le X < 3,\ 1 \le Y < 3]$.
- What is P[A] = P[Y > X]?

Joint Random Variables

- **Marginal PDF**
- Same as the marginal PMF, sometimes we need to find marginal PDF $f_X(x)$ or $f_Y(y)$ from the joint PDF $f_{X,Y}(x,y)$. Therefore, we refer to $f_X(x)$ and $f_Y(y)$ as the *marginal probability density functions* of $f_{X,Y}(x,y)$.

- If X and Y are random variables with joint PDF $f_{X,Y}(x,y)$,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx$$

Joint Random Variables

- **Example:** The joint PDF of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} {}^{5y}/_{4} & -1 \le x \le 1\,, x^2 \le y \le 1 \\ 0 & otherwise \end{cases}$$

- Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.